

PRACTICAL

WEEK 4

2520090046

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1.

The screenshot shows the SQL Developer interface with a script editor containing the following SQL commands:

```
1 DROP DATABASE IF EXISTS bookflow_db;
2 CREATE DATABASE bookflow_db;
3 USE bookflow_db;
4 CREATE TABLE Books (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6     title VARCHAR(200) NOT NULL,
7     author VARCHAR(150) NOT NULL,
8     genre VARCHAR(100),
9     description TEXT,
10    published_year INT
11 );
```

Below the script editor, the 'Output' window displays the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:32:02	DROP DATABASE IF EXISTS bookflow_db	4 row(s) affected	0.047 sec
2	14:32:02	CREATE DATABASE bookflow_db	1 row(s) affected	0.015 sec
3	14:32:02	USE bookflow_db	0 row(s) affected	0.000 sec
4	14:32:33	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(150) NOT NULL, genre VARCHAR(100), description TEXT, published_year INT)	0 row(s) affected	0.032 sec

2.

The screenshot shows the SQL Developer interface with a script editor containing the following SQL commands:

```
12 INSERT INTO Books (title, author, genre, description, published_year)
13 VALUES
14 ('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive alone on Mars.', 2011),
15 ('Dune', 'Frank Herbert', 'Science Fiction', 'A story involving planets, politics, survival and space.', 1965),
16 ('The Hitchhikers Guide to the Galaxy', 'Douglas Adams', 'Science Fiction', 'A humorous adventure across space and alien worlds.', 1979),
17 ('Pride and Prejudice', 'Jane Austen', 'Romance', 'A classic story about relationships and social expectations.', 1813),
18 ('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'A story of ambition, wealth, love and the American dream.', 1925),
19 ('Foundation', 'Isaac Asimov', 'Science Fiction', 'A futuristic story about mathematics, civilization and space.', 1951),
20 ('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 'A fantasy adventure involving Bilbo Baggins and a dangerous journey.', 1937),
21 ('Enders Game', 'Orson Scott Card', 'Science Fiction', 'A young student is trained for an interstellar conflict.', 1985),
22 ('Harry Potter and the Sorcerers Stone', 'J.K. Rowling', 'Fantasy', 'A young wizard begins his magical education.', 1997),
23 ('Twenty Thousand Leagues Under the Seas', 'Jules Verne', 'Adventure', 'A fantastic underwater adventure aboard a mysterious submarine.', 1870);
```

Below the script editor, the 'Output' window displays the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:32:02	DROP DATABASE IF EXISTS bookflow_db	4 row(s) affected	0.047 sec
2	14:32:02	CREATE DATABASE bookflow_db	1 row(s) affected	0.015 sec
3	14:32:02	USE bookflow_db	0 row(s) affected	0.000 sec
4	14:32:33	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(150) NOT NULL, genre VARCHAR(100), description TEXT, published_year INT)	0 row(s) affected	0.032 sec
5	14:35:11	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive alone on Mars.', 2011), ('Dune', 'Frank Herbert', 'Science Fiction', 'A story involving planets, politics, survival and space.', 1965), ('The Hitchhikers Guide to the Galaxy', 'Douglas Adams', 'Science Fiction', 'A humorous adventure across space and alien worlds.', 1979), ('Pride and Prejudice', 'Jane Austen', 'Romance', 'A classic story about relationships and social expectations.', 1813), ('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'A story of ambition, wealth, love and the American dream.', 1925), ('Foundation', 'Isaac Asimov', 'Science Fiction', 'A futuristic story about mathematics, civilization and space.', 1951), ('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 'A fantasy adventure involving Bilbo Baggins and a dangerous journey.', 1937), ('Enders Game', 'Orson Scott Card', 'Science Fiction', 'A young student is trained for an interstellar conflict.', 1985), ('Harry Potter and the Sorcerers Stone', 'J.K. Rowling', 'Fantasy', 'A young wizard begins his magical education.', 1997), ('Twenty Thousand Leagues Under the Seas', 'Jules Verne', 'Adventure', 'A fantastic underwater adventure aboard a mysterious submarine.', 1870);	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

3.

```

24 • ALTER TABLE Books ADD COLUMN embedding JSON;
25
26 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
27 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
28 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
29 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
30 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
31 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
32 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
33 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
34 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
35 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;

```

Output

#	Time	Action	Message	Duration / Fetch
14	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
16	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
17	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
19	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec

4.

```

36 • SELECT
37     book_id, title, author, genre,
38     ROUND(SQRT(
39         POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) - 0.90, 2) +
40         POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) - 0.80, 2) +
41         POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) - 0.20, 2)
42     ), 4) AS l2_distance
43 FROM Books
44 WHERE embedding IS NOT NULL
45 ORDER BY l2_distance ASC
46 LIMIT 3;

```

Result Grid

	book_id	title	author	genre	l2_distance
1		The Martian	Andy Weir	Science Fiction	0
8		Enders Game	Orson Scott Card	Science Fiction	0.0735
2		Dune	Frank Herbert	Science Fiction	0.0866

Result 1 x

Output

#	Time	Action	Message	Duration / Fetch
15	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
16	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
17	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
19	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
20	14:38:18	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSON_EXTRACT(embedding, '\$[0]') AS DECIMAL(10,6)) - 0.90, 2) + POW(CAST(JSON_EXTRACT(embedding, '\$[1]') AS DECIMAL(10,6)) - 0.80, 2) + POW(CAST(JSON_EXTRACT(embedding, '\$[2]') AS DECIMAL(10,6)) - 0.20, 2)), 4) AS l2_distance FROM Books WHERE embedding IS NOT NULL ORDER BY l2_distance ASC LIMIT 3;	3 row(s) returned	0.000 sec / 0.000 sec

5.

```

55      Sqrt(
56          Pow(Cast(Json_Extract(embedding, '$[0]') As Decimal(10,6)), 2) +
57          Pow(Cast(Json_Extract(embedding, '$[1]') As Decimal(10,6)), 2) +
58          Pow(Cast(Json_Extract(embedding, '$[2]') As Decimal(10,6)), 2)
59      ) * Sqrt(Pow(0.90, 2) + Pow(0.80, 2) + Pow(0.20, 2))
60  ), 4
61  ) AS cosine_similarity
62  FROM Books
63  WHERE embedding IS NOT NULL
64  ORDER BY cosine_similarity DESC
65  LIMIT 3;

```

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 2 x

Output

#	Time	Action	Message	Duration / Fetch
16	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
17	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
19	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
20	14:38:18	SELECT book_id, title, author, genre, ROUND(Sqrt(Pow(Cast(Json_Extract(embedding, '\$[0]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[1]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[2]') As Decimal(10,6)), 2)) * Sqrt(Pow(0.90, 2) + Pow(0.80, 2) + Pow(0.20, 2))), 4) AS cosine_similarity FROM Books WHERE embedding IS NOT NULL ORDER BY cosine_similarity DESC LIMIT 3;	3 row(s) returned	0.000 sec / 0.000 sec
21	14:38:56	SELECT book_id, title, author, genre, ROUND(Sqrt(Pow(Cast(Json_Extract(embedding, '\$[0]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[1]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[2]') As Decimal(10,6)), 2)) * Sqrt(Pow(0.90, 2) + Pow(0.80, 2) + Pow(0.20, 2))), 4) AS cosine_similarity FROM Books WHERE embedding IS NOT NULL ORDER BY cosine_similarity DESC LIMIT 3;	3 row(s) returned	0.000 sec / 0.000 sec

6.

```

74      Sqrt(
75          Pow(Cast(Json_Extract(embedding, '$[0]') As Decimal(10,6)), 2) +
76          Pow(Cast(Json_Extract(embedding, '$[1]') As Decimal(10,6)), 2) +
77          Pow(Cast(Json_Extract(embedding, '$[2]') As Decimal(10,6)), 2)
78      ) * Sqrt(Pow(0.90, 2) + Pow(0.80, 2) + Pow(0.20, 2))
79  ), 4
80  ) AS cosine_similarity
81  FROM Books
82  WHERE embedding IS NOT NULL
83  ORDER BY cosine_similarity DESC
84  LIMIT 3;

```

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 3 x

Output

#	Time	Action	Message	Duration / Fetch
17	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
19	14:37:21	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
20	14:38:18	SELECT book_id, title, author, genre, ROUND(Sqrt(Pow(Cast(Json_Extract(embedding, '\$[0]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[1]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[2]') As Decimal(10,6)), 2)) * Sqrt(Pow(0.90, 2) + Pow(0.80, 2) + Pow(0.20, 2))), 4) AS cosine_similarity FROM Books WHERE embedding IS NOT NULL ORDER BY cosine_similarity DESC LIMIT 3;	3 row(s) returned	0.000 sec / 0.000 sec
21	14:38:56	SELECT book_id, title, author, genre, ROUND(Sqrt(Pow(Cast(Json_Extract(embedding, '\$[0]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[1]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[2]') As Decimal(10,6)), 2)) * Sqrt(Pow(0.90, 2) + Pow(0.80, 2) + Pow(0.20, 2))), 4) AS cosine_similarity FROM Books WHERE embedding IS NOT NULL ORDER BY cosine_similarity DESC LIMIT 3;	3 row(s) returned	0.000 sec / 0.000 sec
22	14:46:41	SELECT book_id, title, author, genre, ROUND(Sqrt(Pow(Cast(Json_Extract(embedding, '\$[0]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[1]') As Decimal(10,6)), 2) + Pow(Cast(Json_Extract(embedding, '\$[2]') As Decimal(10,6)), 2)) * Sqrt(Pow(0.90, 2) + Pow(0.80, 2) + Pow(0.20, 2))), 4) AS cosine_similarity FROM Books WHERE embedding IS NOT NULL ORDER BY cosine_similarity DESC LIMIT 3;	3 row(s) returned	0.000 sec / 0.000 sec